

Comparative Forecasting of Global Maritime Port Traffic Using Classical and Advanced Time Series Models

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Abstract—Predicting maritime traffic in ports supports berthing management, crew allocation, congestion reduction, and global supply chain continuity. This paper proposes a time series forecasting framework for international port vessel arrivals using historical port traffic data from 2019–2026. Selected ports include major global hubs such as Singapore, Shanghai, Rotterdam, Busan, Antwerp, Hong Kong, and Japanese industrial ports. Multiple forecasting paradigms are evaluated, including traditional statistical models (ARIMA and Holt-Winters), multivariate approaches (VAR), and modern techniques (Facebook Prophet and XGBoost). Exploratory analysis reveals traffic trends, weekly seasonality, disruption events, and generally low global inter-port correlations with stronger regional dependencies. Results show that no single model dominates all ports. ARIMA performs best for smooth low-volatility ports, Prophet for seasonal ports, and XGBoost for volatile traffic patterns. The findings highlight the importance of adaptive forecasting strategies under uncertain global trade conditions.

Index Terms—Time Series Forecasting, Maritime Analytics, Port Traffic, ARIMA, Prophet, XGBoost, Supply Chain Intelligence

I. INTRODUCTION

Global seaports are critical nodes in international trade networks. Congestion, bottlenecks, and poor planning at ports can disrupt supply chains worldwide. Accurate short-term forecasting of vessel arrivals helps optimize berth allocation, manpower scheduling, storage planning, and inland logistics.

Traditional forecasting approaches such as ARIMA remain widely used because of their robustness. However, modern maritime traffic data often contain nonlinear shocks, seasonal effects, and regional dependencies. This motivates the use of newer forecasting methods such as Prophet and machine learning models.

This study compares multiple forecasting approaches across major global ports to determine which models perform best under different traffic conditions.

The key contributions of this study are:

- Comparative review of classical and modern forecasting methods on global port traffic data.
- Exploratory analysis of trends, seasonality, disruptions, and correlations.

- Identification of suitable models for stable, seasonal, and volatile ports.
- Practical recommendations for maritime planning systems.

II. LITERATURE REVIEW

Time series forecasting has long been studied in transportation and logistics systems.

Box and Jenkins introduced ARIMA, one of the most widely used statistical forecasting methods. Holt-Winters exponential smoothing is effective for level, trend, and seasonal series.

Taylor and Letham proposed Prophet, an additive forecasting model capable of handling trend changes and multiple seasonalities.

XGBoost is a gradient boosting framework capable of learning nonlinear relationships and lag dependencies. It has shown strong performance in structured forecasting tasks.

Recent maritime forecasting studies focus on vessel arrival prediction, congestion estimation, and throughput forecasting. However, many studies focus on single ports or single algorithms. Cross-port benchmarking remains relatively underexplored.

III. DATASET AND PREPROCESSING

A. Dataset

The dataset contains daily vessel arrival counts from 2019–2026 for major international ports, including:

- Singapore
- Shanghai
- Rotterdam
- Antwerp
- Busan
- Hong Kong
- Ningbo
- Japanese industrial ports: Nagoya, Kobe, Chiba, Mizushima

B. Preprocessing Steps

The following preprocessing steps were performed:

- Date parsing and chronological sorting
- Missing value treatment
- Daily frequency alignment
- Port-wise segmentation
- Chronological train-test split
- Lag feature generation for machine learning models
- Stationarity checks for ARIMA

C. Exploratory Findings

Initial analysis revealed:

- Singapore as the highest-volume relatively stable port
- Greater variability in Shanghai traffic
- Strong similarities between Rotterdam and Antwerp
- Weekly cycles in some ports
- Traffic disruptions during COVID-19, Suez Canal blockage, and Red Sea tensions

IV. METHODOLOGY

A. Classical Models

ARIMA: Autoregressive Integrated Moving Average model for linear forecasting after stationarity adjustments.

Holt-Winters: Triple exponential smoothing for level, trend, and seasonality.

B. Multivariate Model

VAR: Vector Autoregression used for regionally correlated ports to model interactions among multiple time series.

C. Advanced Models

Prophet: Automated trend and seasonality model with changepoint detection.

XGBoost: Supervised learning model trained using lag-based features such as:

- Lag-1 traffic
- Lag-7 traffic
- Rolling averages
- Regional leader lag values
- Calendar variables

D. Evaluation Metrics

Models were evaluated using RMSE and MAPE:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (2)$$

RMSE penalizes large errors, while MAPE enables scale-independent comparison across ports.

TABLE I
OBSERVED MODEL STRENGTHS

Model	Best Use Case
ARIMA	Smooth low-volatility ports
Holt-Winters	Stable seasonal series
Prophet	Strong weekly seasonality and trend shifts
XGBoost	Nonlinear and volatile traffic
VAR	Regionally coupled ports

V. RESULTS AND DISCUSSION

A. Comparative Performance

No single model consistently outperformed others across all ports.

B. Key Findings

- Prophet performed strongly on seasonal ports.
- XGBoost performed well on Japanese industrial ports.
- ARIMA worked effectively on smoother traffic flows.
- Holt-Winters was stable but slower to react to shocks.
- Low global correlation suggests many ports can be modeled independently.
- Regional clusters benefit from VAR models.

C. Interpretation

Port traffic behavior varies by geography, economic role, and operational structure. Therefore, adaptive model selection is preferred over a universal forecasting approach.

VI. REAL-WORLD APPLICATIONS

Accurate port traffic forecasting supports:

- Berth allocation
- Crane and labor scheduling
- Yard capacity planning
- Congestion mitigation
- Route and fuel optimization
- Early disruption warning systems

VII. LIMITATIONS

This study has several limitations:

- Exogenous variables such as weather and fuel prices were not included.
- Abrupt geopolitical events remain difficult to predict.
- Deep learning models were not extensively tuned.
- Port traffic data can be noisy.
- Long-term forecasting was not the primary focus.

VIII. CONCLUSION AND FUTURE WORK

This study benchmarked classical, multivariate, and machine learning approaches for global port traffic forecasting. Results confirm that no single model fits all ports. ARIMA remains strong for stable series, Prophet excels in seasonal environments, and XGBoost performs well on volatile traffic.

Future work may include:

- Deep learning models such as LSTM and GRU
- Transformer-based temporal forecasting
- Weather and macroeconomic regressors

- Real-time AIS vessel stream integration
- Ensemble forecasting systems

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